Lab 11 OTP – Web Form and Table Design

1. Duplicate the *Registration.aspx* web form as shown in Figure 1. Rename it as regOTP.aspx.

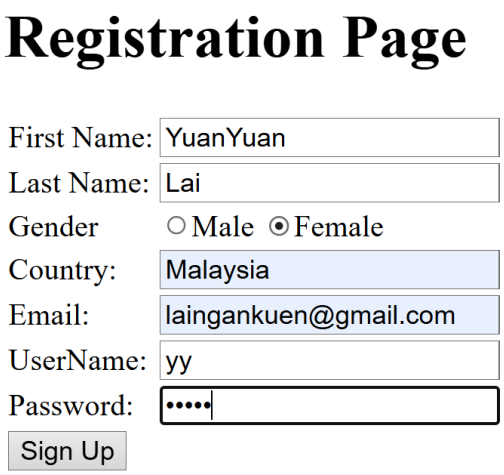


Figure 1: Registration Form

2. Copy the c# script from *Registration.aspx.cs.* Modify it as depicted in Figure 2 and 3.

```
// ===== OTP PART =====
string otp = OtpHelper.GenerateNumericOtp();
string secret = ConfigurationManager.AppSettings["OtpSecret"];
string otpHash = OtpHelper.HmacOtp(otp, secret);

// Save OTP into EmailOtp table
OtpHelper.SaveOtpToDb(newUserId, otpHash, DateTime.UtcNow.AddMinutes(10));

// Send OTP email
EmailHelper.SendOtpEmail(email.Text, otp);

// Redirect to OTP verification page
Response.Redirect("VerifyOtp.aspx?userId=" + newUserId);
```

Figure 2: OTP scripts

```
using System;
using System.Collections;
using System.Collections.Generic;
using System.Configuration;
using System.Data.SqlClient;
using System.Linq;
using System.Security.Cryptography;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using Twilio.TwiML.Voice;
```

Figure 3: C# namespaces

3. Create a web form called *VerifyOTP.aspx.cs* as shown in Figure 4.

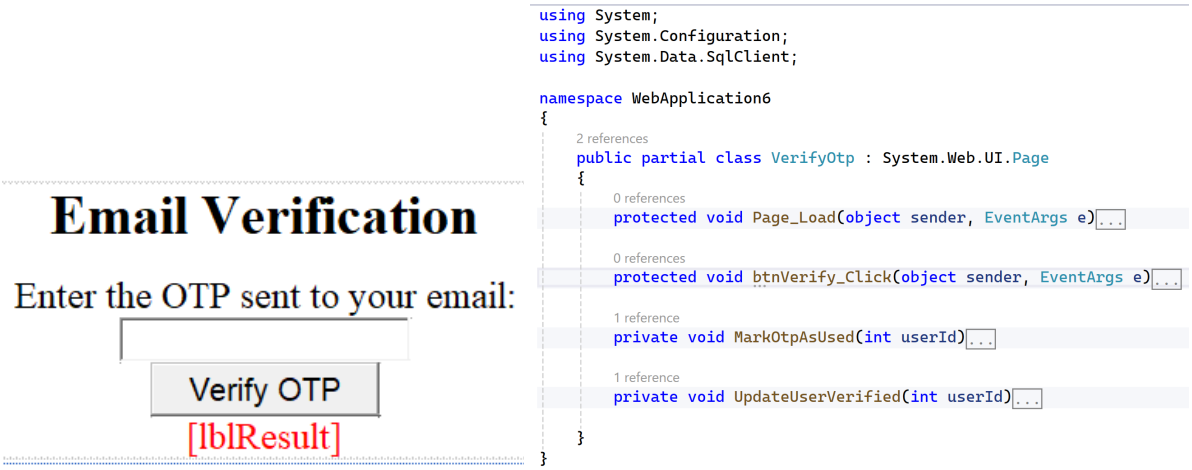


Figure 4: Email verification

4. User will receive the OTP via registered email (see Figure 5).

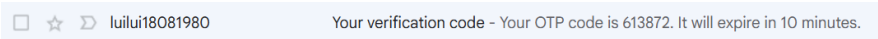


Figure 5: Email Notification

5. User must fill in the received OTP before the code is expired (see Figure 6).

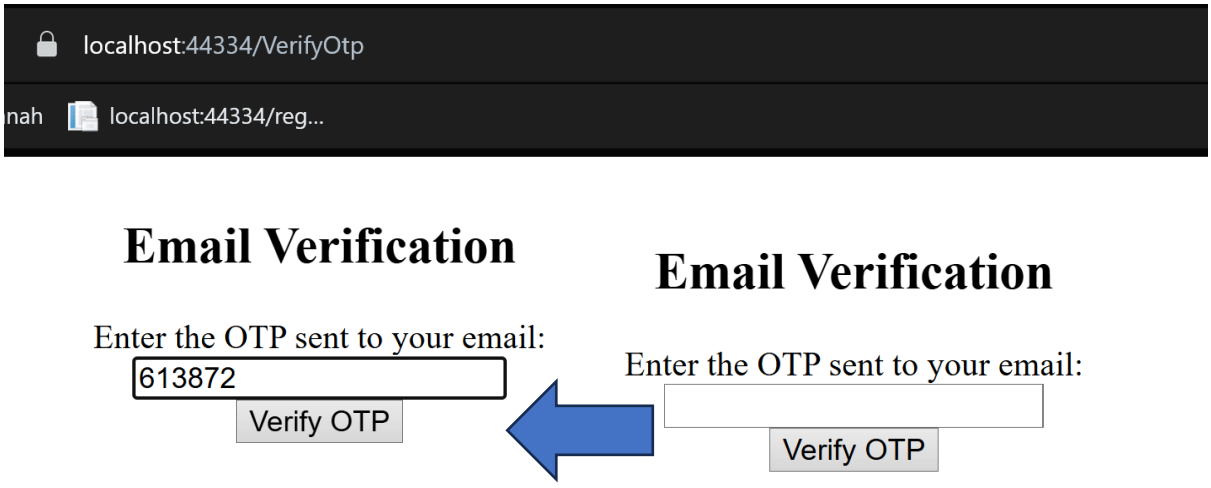


Figure 6: Email Verification Process

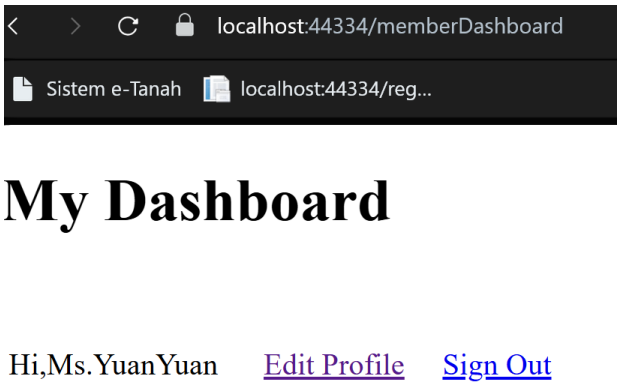
6. For successful registration, you may get the following outputs as illustrated in Figure 7:-

   i) Login Page (see Figure 7(a))

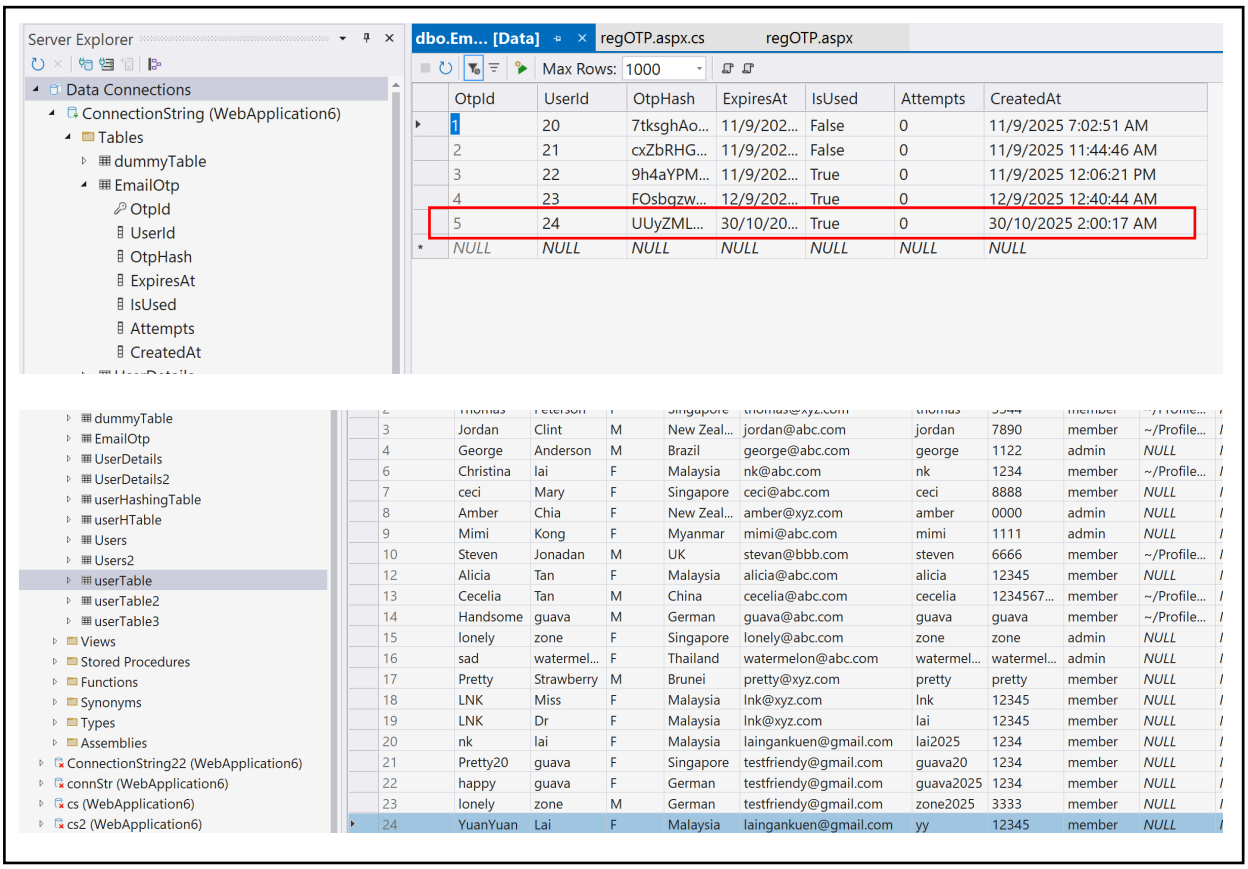
   ii) Member Dashboard (see Figure 7(b))

   ii) Database (see Figure 7(c))

(a)



(b)



(c)

Figure 7:Expected Outputs

Remark:

1) Gmail account setup for sending the OTP to new user.

2) You may try other alternative approach to implement the OTP mechanism.